IAP
INSTITUTE OF ADVANCED PHYSICS
COLLEGIATE STUDIES IN NATURAL PHILOSOPHY
MONOGRAPH 01
MASTERCLASS SERIES

SENIOR PHYSICS · ELECTRODYNAMICS · MASTERCLASS EDITION

Alternating Current

Class XII · A deep visual guide to phasors, reactance, resonance, power and transformers

October 11, 2026

Abstract

Alternating current is not simply "current that changes direction." It is a continuously evolving electrical state in which voltage, current, stored electric energy and stored magnetic energy exchange in a precise rhythm. These notes build the chapter from the rotating-vector picture, so every formula has a physical meaning.

1 Learning goals

- ☐ Interpret sinusoidal AC with waveforms and phasors.
- ☐ Explain resistor, inductor and capacitor phase relations from first principles.
- ☐ Solve series LCR, resonance, power-factor and transformer problems.

> **How to use these notes.**
>
> Preview the diagram first, cover each worked solution and attempt it yourself, then answer the retrieval questions without looking back. Revisit any question you cannot explain aloud.

2 The big picture: what alternates and why?

A direct current keeps one direction. An alternating current changes both **magnitude and direction periodically**. Power stations generate AC because electromagnetic induction

naturally produces a sinusoid, transformers can then change its voltage efficiently, and high-voltage transmission greatly reduces heating loss.

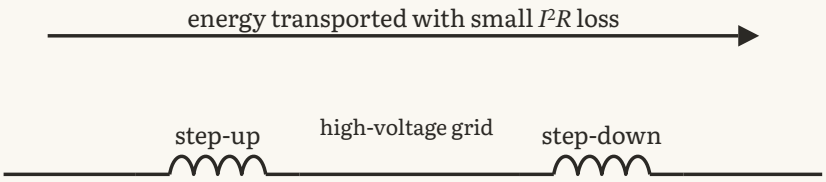


Figure 1: The logic of the AC power system: generate, transform, transmit and transform again.

> **One idea to keep throughout.**
>
> The source sets the **voltage frequency**. The circuit elements decide the **current amplitude** and its **phase** relative to voltage.

3 Sinusoidal voltage and current

For a sinusoidal source, instantaneous voltage and current are written

$$v(t) = V_m \sin(\omega t), \qquad i(t) = I_m \sin(\omega t + \phi) \tag{1}$$

Symbol	Meaning	SI unit
V_m, I_m	peak (maximum) values	V, A
ω	angular frequency $= 2\pi f$	rad s^{-1}
f	cycles per second	Hz
T	time for one cycle $= 1/f$	s
ϕ	phase of current relative to voltage	rad or degree

Graph insight.

Two complete cycles. Horizontal separation measures phase difference.

Reading a sinusoid deeply.

At a zero crossing the quantity is momentarily zero but changing fastest. At a crest its value is greatest but its rate of change is zero. This distinction explains inductor and capacitor phase relations.

4 Mean and RMS values

The average of a sine wave over a complete cycle is zero, which does **not** mean AC delivers no energy. Heating depends on i^2, which stays positive. The RMS current is the DC current producing the same heat in a resistor.

$$I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} = \frac{I_m}{\sqrt{2}}, \qquad V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \tag{2}$$

Household rating.

A 230 V AC supply means **230 V RMS**, not peak. Its peak is $V_m = 230\sqrt{2} \approx 325$ V.

1. **Step 1:** Start with $i = I_m \sin \omega t$.
2. **Step 2:** $I_{rms}{}^2 = 1/T \int_0{}^T I_m{}^2 \sin{}^2 \omega t \, dt$
3. **Step 3:** Over a full cycle, the mean of $\sin{}^2$ is $1/2$.
4. **Step 4:** $I_{rms} = I_m/\sqrt{2}$

5 The phasor method

A phasor is a rotating vector. Its vertical projection produces the instantaneous sinusoid. Phasors convert differential relations into geometry: vector lengths represent amplitudes and angular separation represents phase.

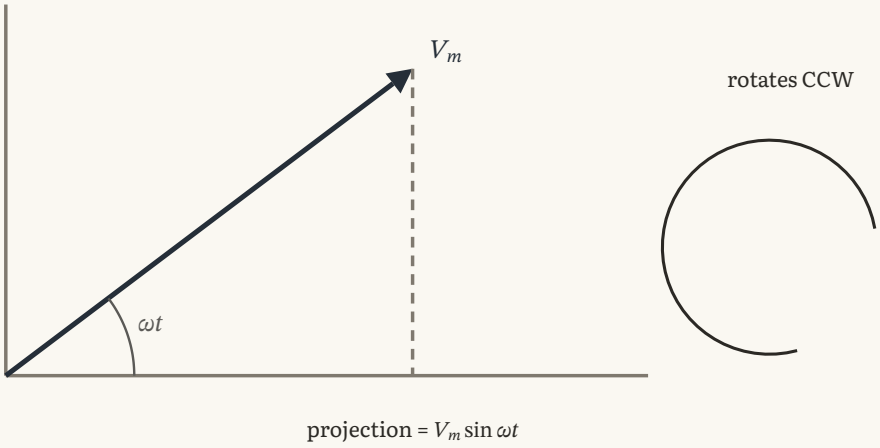


Figure 2: A sinusoid is the shadow of uniform circular motion. The phasor itself is not a physical rotating wire.

6 Pure resistor: voltage and current in phase

A resistor stores no energy. At each instant Ohm's law applies, so current rises, crosses zero and falls exactly with voltage.

$$v = V_m \sin \omega t, \quad i = \frac{v}{R} = I_m \sin \omega t, \quad I_m = V_m / R \tag{3}$$

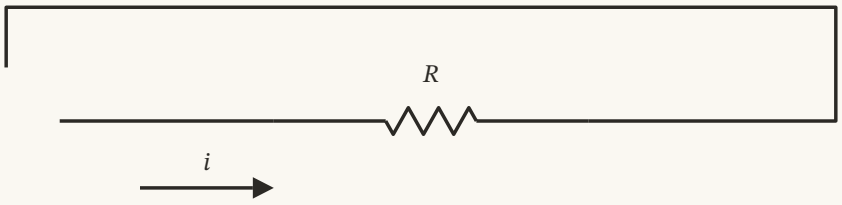


Figure 3: Pure resistive AC circuit.

$$P_{av} = V_{rms} I_{rms} = I_{rms}^2 R = \frac{V_{rms}^2}{R} \tag{4}$$

7 Pure inductor: current lags by 90°

An inductor opposes a **change** in current by producing self-induced emf. From $v = L\,di/dt$, integration shows that current reaches its maximum one-quarter cycle after voltage.

$$i = I_m \sin(\omega t - \pi/2), \qquad X_L = \omega L, \qquad I_m = \frac{V_m}{X_L} \tag{5}$$

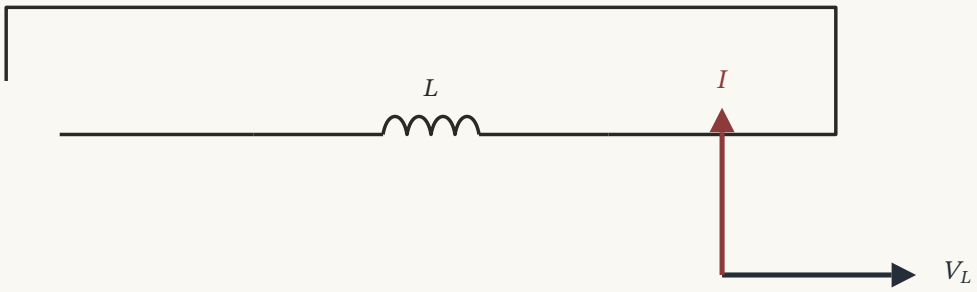


Figure 4: Circuit and phasors: V_L leads I by 90°.

> **Frequency response.**
>
> At high frequency, current must change rapidly, producing a large back emf. Therefore $X_L = 2\pi f L$ increases with frequency: an inductor passes low frequency more easily than high frequency.

During one quarter-cycle the source stores magnetic energy $U_B = 1/2 L i^2$; during the next it receives that energy back. Hence an ideal inductor has zero average power.

8 Pure capacitor: current leads by 90°

A capacitor draws current while its charge—and therefore voltage—is changing. Since $i = C\,dv/dt$, current is maximum when voltage crosses zero and zero when voltage is at an extreme.

$$i = I_m \sin(\omega t + \pi/2), \qquad X_C = \frac{1}{\omega C}, \qquad I_m = \frac{V_m}{X_C} \tag{6}$$

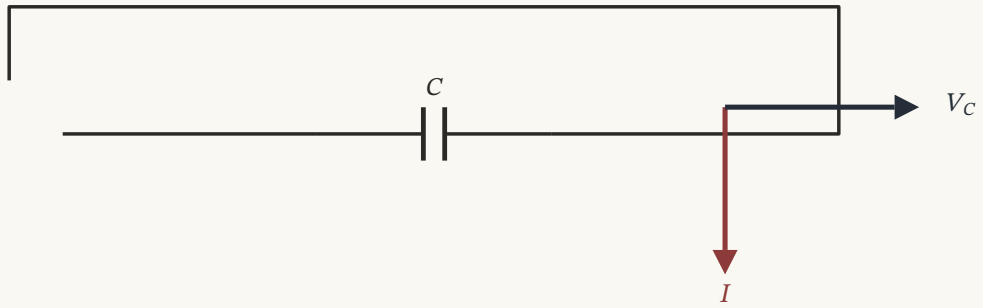


Figure 5: Circuit and phasors: I leads V_C by 90°.

> **Frequency response.**
>
> At high frequency there is little time to accumulate charge, so a small voltage drives appreciable current. Thus $X_C = 1/(2\pi fC)$ decreases with frequency: a capacitor blocks DC but passes rapidly varying signals.

9 Series LCR circuit

The same current flows through all three elements. Use current as the reference phasor. V_R lies along I, V_L points upward, and V_C points downward. Because opposite phasors subtract, voltages must **not** be added arithmetically.

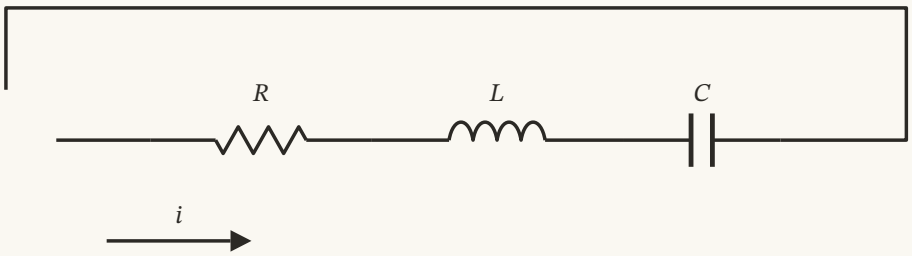


Figure 6: A series LCR circuit driven by a sinusoidal source.

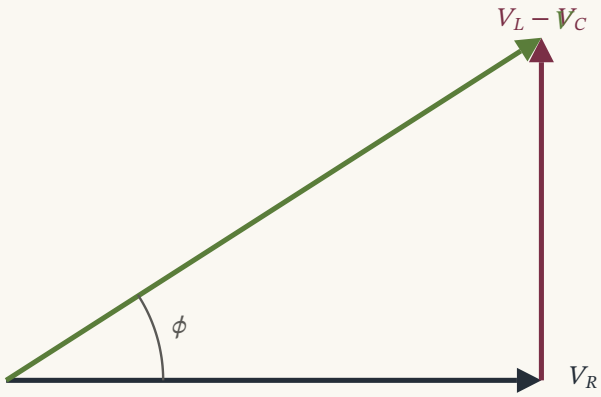


Figure 7: Voltage triangle for the net inductive case $X_L > X_C$.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \qquad I_{rms} = \frac{V_{rms}}{Z} \tag{7}$$

$$\tan\phi = \frac{X_L - X_C}{R}, \qquad \cos\phi = \frac{R}{Z} \tag{8}$$

Condition	Nature	Current relative to source voltage

$X_L > X_C$	inductive	lags
$X_C > X_L$	capacitive	leads
$X_L = X_C$	resonant / resistive	in phase

10 Resonance

At resonance the inductive and capacitive reactances cancel. Impedance becomes minimum, current becomes maximum and the source only replaces energy lost in resistance. Large internal voltages may still appear across L and C.

$$\omega_0 L = \frac{1}{\omega_0 C} \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}} \tag{9}$$

> **Graph insight.**
>
> Resonance curves. Lower resistance gives greater selectivity and a sharper peak.

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}, \qquad \Delta f = \frac{f_0}{Q} \tag{10}$$

> **Why tuning works.**
>
> A radio's variable capacitor changes f_0. At the desired station frequency, current response is large; nearby frequencies are suppressed. High Q means good selectivity.

11 AC power and power factor

Instantaneous power $p = vi$ fluctuates. Averaging over a cycle gives

$$P_{av} = V_{rms} I_{rms} \cos\phi \tag{11}$$

The factor $\cos\phi$ tells what fraction of apparent power is converted to net useful energy. Inductors and capacitors exchange reactive energy with the source; resistance dissipates real energy.

Quantity	Expression	Unit
Real (active) power	$P = VI\cos\phi$	W
Reactive power	$Q = VI\sin\phi$	var
Apparent power	$S = VI$	VA

Wattless current does not mean zero current.

For a pure L or C, $\phi = 90°$ and average power is zero, but instantaneous current and energy exchange are real. "Wattless" means no **net energy per cycle**.

Why industries correct power factor.

For fixed useful power, $I = P/(V\cos\phi)$. A low power factor requires larger current, causing larger I^2R losses and thicker cables. Capacitor banks compensate inductive loads.

12 Transformer

A transformer uses **mutual induction** to transfer AC power between two coils on a common laminated iron core. The changing primary current creates changing core flux, inducing emf in both windings.

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}, \qquad V_p I_p = V_s I_s, \qquad \frac{I_s}{I_p} = \frac{N_p}{N_s} \tag{12}$$

12.1 Transformer losses and their reduction

- **Copper loss:** I^2R heating; use thick, low-resistance copper.
- **Eddy-current loss:** currents induced in the core; use thin insulated laminations.
- **Hysteresis loss:** repeated magnetisation; use soft iron or silicon steel.
- **Flux leakage:** incomplete linkage; use a closed, high-permeability core.

Important: A transformer does not work on steady DC. DC gives only a brief transient at switching and can overheat the primary because inductive reactance disappears.